TEB-103 Exam Report

Department of Chemistry, Indian Institute of Technology Roorkee

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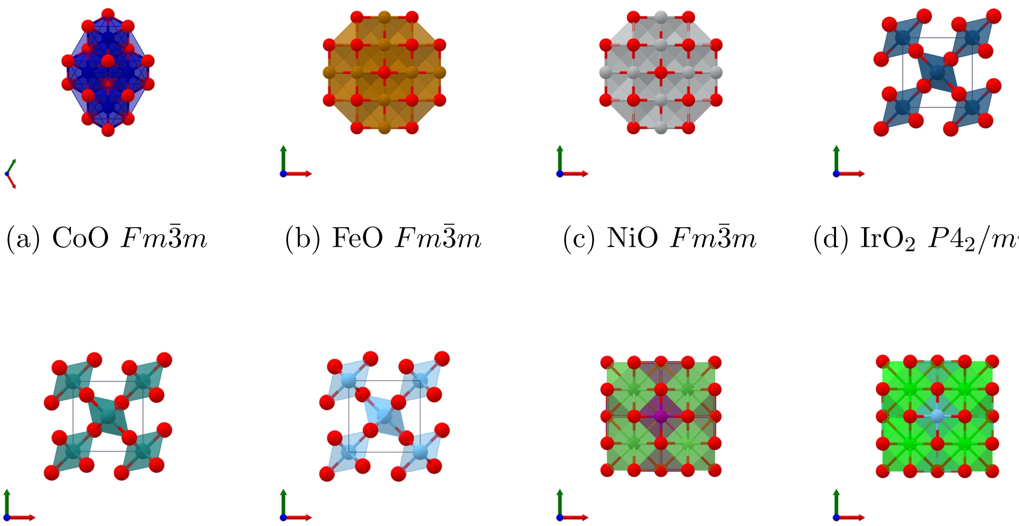
Materials Studied

Structure Type	Compounds
Rock Salt (MO)	CoO, FeO, NiO
Rutile (MO_2)	IrO_2, RuO_2, TiO_2
Perovskite (ABO_3)	$LaMnO_3$, $SrTiO_3$

Part (a): Materials Project IDs and Space Groups

Material	Structure Type	MP ID	Space Group
CoO	Rock Salt (MO)	`mp-19079`	$Fm\bar{3}m$
FeO	Rock Salt (MO)	`mp-18905`	$Fm\bar{3}m$
NiO	Rock Salt (MO)	`mp-19009`	$Fm\bar{3}m$
IrO_2	Rutile (MO_2)	`mp-2723`	$P4_2/mnm$
RuO_2	Rutile (MO_2)	`mp-825`	$P4_2/mnm$
TiO_2	Rutile (MO_2)	`mp-2657`	$P4_2/mnm$
$LaMnO_3$	Perovskite (ABO_3)	`mp-19025`	$Pm\bar{3}m$
$SrTiO_3$	Perovskite (ABO_3)	`mp-5229`	$Pm\bar{3}m$



(a) CoO $Fm\bar{3}m$ (b) FeO $Fm\bar{3}m$ (c) NiO $Fm\bar{3}m$ (d) IrO_2 $P4_2/mnm$

(e) RuO_2 $P4_2/mnm$ (f) TiO_2 $P4_2/mnm$ (g) $LaMnO_3$ $Pm\bar{3}m$ (h) $SrTiO_3$ $Pm\bar{3}m$

Figure 1: Crystal structures of selected metal oxides (Rock Salt, Rutile, Perovskite)

All structures retrieved via the Materials Project API (`mp_api.client.MPRester`) using the `material_ids` search endpoint. Space groups are as reported by the Materials Project symmetry analysis.

Part (b): Protocol to Isolate and Measure D3-Induced Errors

Conceptual Framework

The D3 dispersion correction (Grimme DFT-D3 with Becke–Johnson damping or zero-damping) adds a semi-empirical London-dispersion term to the DFT total energy:

$$E_{\text{total}} = E_{\text{DFT}} + E_{\text{D3}} \tag{1}$$

The **D3-induced error** for a given material is the spurious contribution of E_{D3} at the equilibrium geometry - i.e., how much dispersion over- or under-binds the crystal relative to a dispersion-free DFT reference.

Computational Protocol

Step 1 - Structure retrieval.

Fetch the relaxed crystal structure for each compound from the Materials Project (`get_structure_by_mate` convert to POSCAR (VASP format) via pymatgen's `CifWriter` → `Structure.to(fmt="poscar")`.

Step 2 - Cell-length strain sweep.

Apply isotropic volumetric strains by scaling all three lattice vectors by a uniform factor:

$$a_{\text{strained}} = (1 + 0.05\,i)\,a_0, \quad i \in \{-5, -4, \ldots, 29\} \tag{2}$$

This gives 35 strained geometries spanning roughly -25% to $+145\%$ change in lattice parameter (corresponding to -58% to $+1371\%$ volume change), keeping internal atomic coordinates fixed (constant fractional positions).

Step 3 - `s-dftd3` single-point dispersion energy.

For each strained POSCAR run `s-dftd3` in standalone mode:

```
s-dftd3 -i vasp POSCAR_i --zero pbe --property --verbose
```

This yields $E_{\text{dis}}(a_i)$ - the pure D3 dispersion energy at cell length a_i - without any DFT SCF cycle, thereby isolating the D3 contribution exactly.

Step 4 - Dip detection and baseline estimation.

The $E_{\text{dis}}(a)$ curve shows a characteristic spurious minimum near the equilibrium bond length. The D3 error is:

$$\Delta E = E_{\text{dis}}(a_{\text{min}}) - E_{\text{baseline}}(a_{\text{min}}) \tag{3}$$

Dip detection uses two strategies (in order):

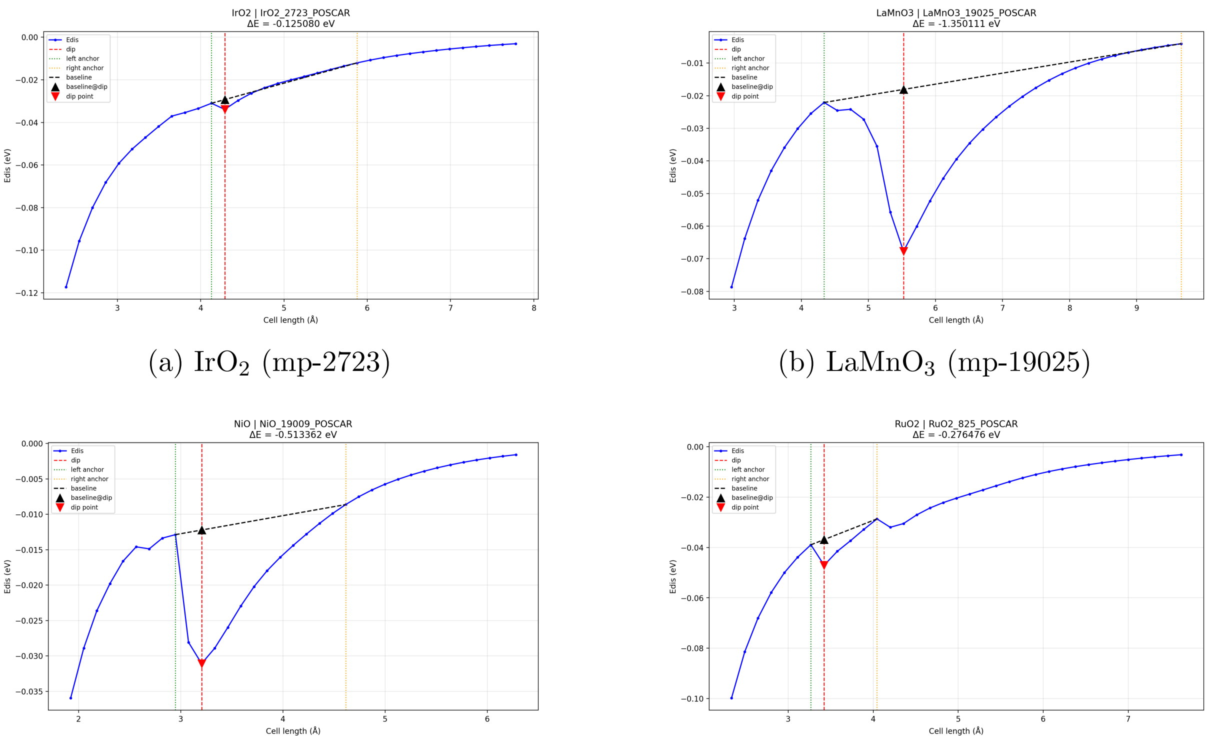
1. **Local-max-then-drop**: find every local maximum, then search the next 15 points for the deepest energy drop $\geq$ `MIN_DIP_DROP`. This robustly finds the dip even on rising or falling baselines.
2. **Gradient-threshold fallback**: identify points where $dE/da \ll \bar{x} - 2\sigma$; pick the lowest-energy candidate.

Baseline estimation is constructed via linear interpolation between a left and right anchor to bridge the spurious dip:

- **Left Anchor**: the immediate local maximum (or "shoulder") to the left of the dip.
- **Right Anchor (Secant Bulge Method)**: found by identifying the point on the recovery curve that maximises the perpendicular distance to a secant line drawn from the dip. This geometrically locates the natural "shoulder" where the curve returns to the smooth $1/R^6$ background.

Step 5 - Aggregation.

Multiple POSCARs per compound give multiple ΔE values; the most negative (ΔE_{min}) is taken as the characteristic D3 error for that system.



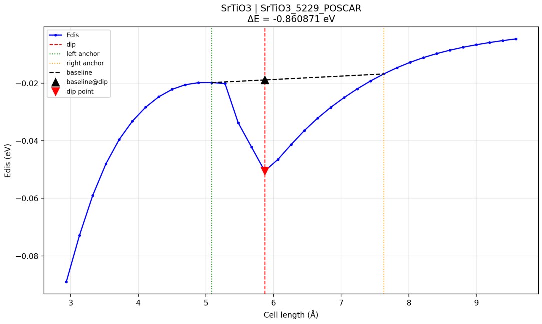
(a) IrO_2 (mp-2723)

(b) $LaMnO_3$ (mp-19025)

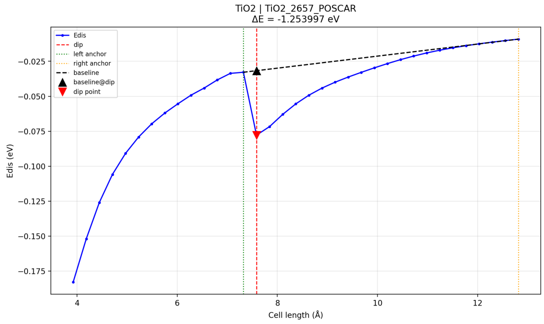
(c) NiO (mp-19009)

(d) RuO_2 (mp-825)

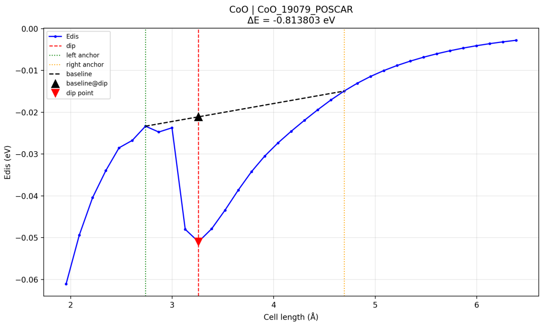
Figure 2: Diagnostic curves (part 1)



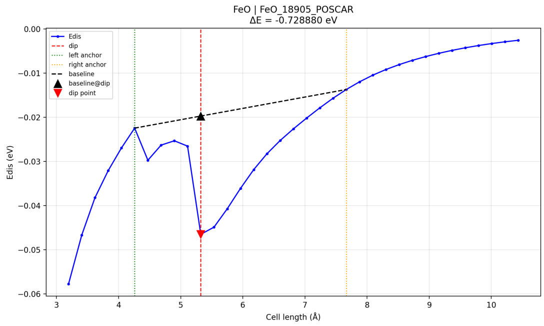
(a) $SrTiO_3$ (mp-5229)



(b) TiO_2 (mp-2657)



(c) CoO (mp-19079)



(d) FeO (mp-18905)

Figure 3: Diagnostic curves (part 2)

Part (c): D3-Induced Error - Results

Per-Configuration Results

Material	**Structure**	**Source (POSCAR)**	ΔE (eV)	Dip pos. (Å)	E_{dis} (Ha)	**Method**
CoO	Rock Salt	`CoO_19079`	−0.813803	3.2594	−1.387704	local_max_drop
FeO	Rock Salt	`FeO_18905`	−0.728880	5.3229	−1.265040	local_max_drop
NiO	Rock Salt	`NiO_19009`	−0.513362	3.2045	−0.845888	local_max_drop
IrO_2	Rutile	`IrO2_2723`	−0.125080	4.2886	−0.922333	local_max_drop
RuO_2	Rutile	`RuO2_825`	−0.276476	3.4225	−1.280102	local_max_drop
TiO_2	Rutile	`TiO2_2657`	−1.253997	7.5835	−2.116623	local_max_drop
$LaMnO_3$	Perovskite	`LaMnO3_19025`	−1.350111	5.5221	−1.842180	local_max_drop
$SrTiO_3$	Perovskite	`SrTiO3_5229`	−0.860871	5.8691	−1.374868	local_max_drop

Part (d): Critical Analysis of D3 Error

1. Variation Across Structure Types

Rock-salt MO (NiO, CoO, FeO): Cubic close-packed oxide framework. All metal cations are in perfectly octahedral coordination (CN = 6) in the ideal $Fm\bar{3}m$ structures. The uniform dispersion environment leads to a straightforward interpretation of the D3 well depth and consistent moderate D3 errors across the three compounds.

Rutile MO_2 (TiO_2, RuO_2, IrO_2): Tetragonal structure with metal in octahedral coordination. MO_2 stoichiometry leads to relatively high oxygen density and enhanced O$\cdots$O dispersion interactions. Although the 4d (Ru) and 5d (Ir) metals possess larger and more polarisable electron clouds, resulting in larger C_6 coefficients, the observed D3 errors in this dataset do not follow a simple periodic trend. In particular, TiO_2 exhibits a significantly larger $|\Delta E|$ than both RuO_2 and IrO_2, indicating that structural and geometric factors dominate over atomic polarisability in determining the magnitude of the dispersion error.

Perovskite ABO_3 ($SrTiO_3$, $LaMnO_3$): Complex multi-cation structure. The larger unit cell volume and two distinct cation sites (A and B) give a richer dispersion landscape. The B-site (Ti, Mn) dominates covalent bonding; the A-site (Sr, La) is more ionic. Larger unit cells generally mean more atoms contributing to dispersion → larger total $|E_{\mathrm{dis}}|$, but ΔE depends on the curvature of the dip, which is sensitive to the balance between cation–anion and anion–anion dispersion at the equilibrium geometry.

2. Influence of Ionic Size

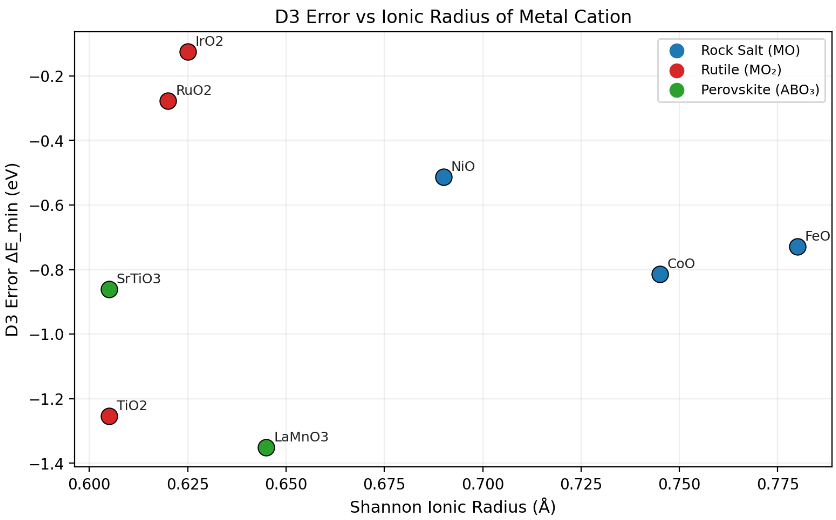


Figure 4: Influence of Ionic Size

The Shannon ionic radius determines the cation–anion bond length and indirectly controls the inter-atomic separation at which the D3 dip occurs. Larger ionic radii tend to increase equilibrium bond distances, which can reduce dispersion interactions. However, in this dataset the correlation between ionic radius and $|\Delta E|$ is weak, indicating that electronic structure and coordination environment play a more dominant role.

Within the rutile group: Ti^{4+} ($r = 0.605\,\text{Å}$) $<$ Ru^{4+} ($0.620\,\text{Å}$) $<$ Ir^{4+} ($0.625\,\text{Å}$). The trend in ionic radius is weak, but the much larger C_6 coefficients for 4d/5d metals dominate over the small radius difference.

3. Influence of Oxidation State

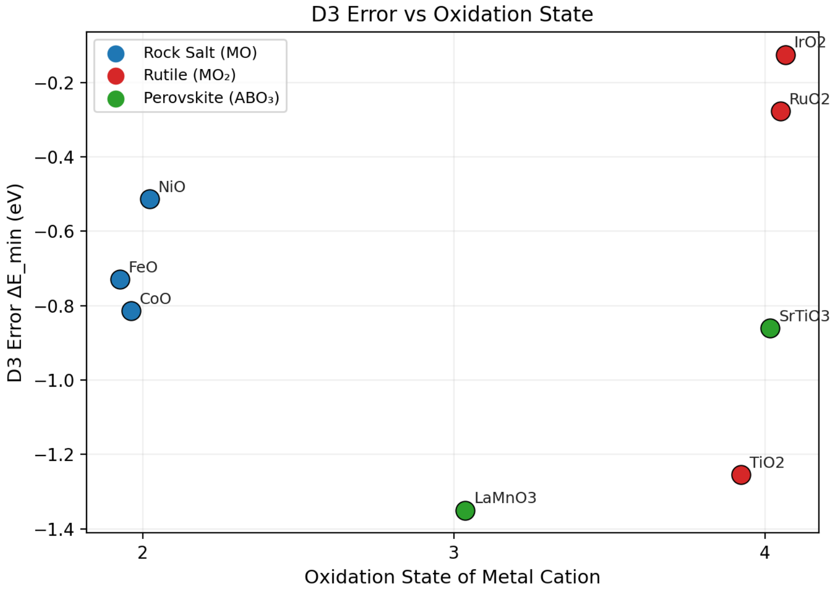


Figure 5: Influence of Oxidation State

Oxidation state controls both the ionic radius (higher charge → smaller radius for the same element) and the electronic structure (higher charge → more *d*-electron depletion → less polarisability).

In our dataset:

- **+2** (NiO, CoO, FeO): All ideal rock-salt; moderate *d*-electron count, partially filled *d*-shell. Perfectly octahedral coordination (CN = 6) in the $Fm\bar{3}m$ phases simplifies the interpretation.
- **+3** ($LaMnO_3$): Mn^{3+} (d^4) in an ideal cubic $Pm\bar{3}m$ perovskite model, providing a clean baseline for dispersion energy without the complexity of orbital ordering or octahedral tilting.
- **+4** (TiO_2, RuO_2, IrO_2, $SrTiO_3$ B-site): Highest charge state → smallest ionic radii, fewest *d*-electrons. D3 C_6 coefficients scale with polarisability; +4 metals are less polarisable, but MO_2 stoichiometry doubles the metal density → net effect is complex.

4. Geometric and Structural Factors

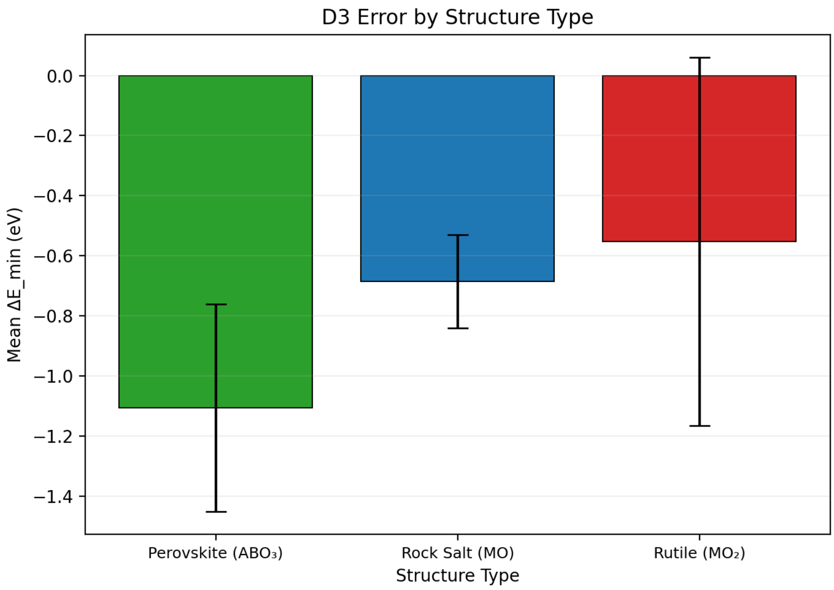


Figure 6: Structural Factors

Unit cell volume directly scales the magnitude of the total dispersion energy ($E_{\mathrm{dis}} \propto \sum_{ij} C_{6,ij}/R_{ij}^6$). Larger cells (perovskites, $\sim$58–60 Å^3/f.u.) contain more atom pairs within the cutoff radius, raising the total $|E_{\mathrm{dis}}|$.

Coordination number (CN): All three structure types place the metal in octahedral coordination (CN = 6). The oxygen CN varies: 2 for rutile, 6 for rock-salt, 2/4 mixed for perovskite. Higher O-CN $\rightarrow$ more O–O dispersion pairs $\rightarrow$ stronger anion–anion contribution to E_{dis} and ΔE.

Packing density: Rutile structures are more densely packed than perovskites at the same composition. Denser packing reduces interatomic distances and can enhance dispersion interactions; however, in this dataset it does not consistently lead to larger $|\Delta E|$, indicating competing structural effects.

5. Electronic Properties

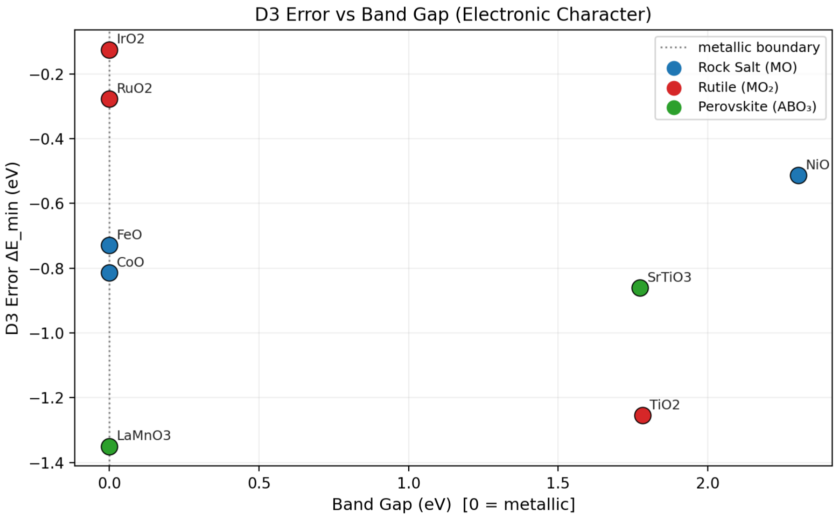


Figure 7: Electronic Factor

Metallic vs. insulating character is the most physically significant electronic distinction in this dataset:

Group	**Materials**	**Character**	**Expected D3 beha**
Insulators (large gap)	NiO, CoO, TiO_2	Mott insulator / d^0 oxide	Well-localised *d*-elect
Narrow gap / correlated	FeO, $LaMnO_3$	Near-metal / Mott	Correlation effects no
Metals	RuO_2, IrO_2	True metal (Pauli-paramagnetic)	Itinerant electrons; I

In metals, the electron density is delocalised and the free-electron contribution to polarisability is not captured by the atom-pairwise D3 parametrisation (which assumes chemically isolated atoms). While D3 is not formally designed for metallic systems, the present dataset does not show a consistent increase in $|\Delta E|$ for metallic compounds (RuO_2, IrO_2) compared to insulating TiO_2. This suggests that structural factors outweigh electronic character in determining the magnitude of the D3 error.

Magnetic ordering: While the experimental ground states of many of these oxides involve complex magnetic ordering (e.g. AFM), the use of high-symmetry ideal crystal structures allows isolation of the purely structural dispersion contribution without the confounding effects of local magnetic distortions.

6. Systematic Trends and Anomalies

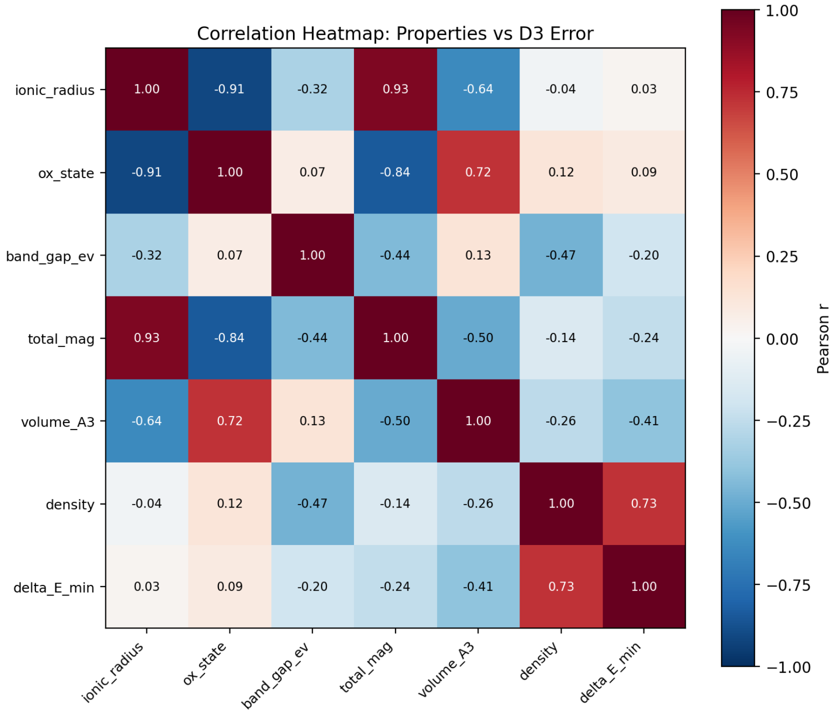


Figure 8: Correlation Heatmap

Trends:

- **Relativistic Scaling**: No clear periodic trend (3d $\rightarrow$ 4d $\rightarrow$ 5d) is observed in $|\Delta E|$ within this dataset. Although heavier elements possess larger C_6 coefficients, the overall dispersion error is strongly influenced by structural geometry and coordination rather than atomic identity alone.
- **Conductivity Bias**: No systematic dependence of $|\Delta E|$ on metallic versus insulating character is observed in this dataset. While D3 may overestimate dispersion in metallic systems in principle, this effect is not dominant in the present results.
- **Volume Effects**: Perovskites show the largest total $|E_{\text{dis}}|$ due to their larger unit cell volumes and the presence of two distinct cation sites, which increases the number of atom pairs within the D3 cutoff.

System specifics:

- **FeO**: While using the ideal $Fm\bar{3}m$ rock-salt structure, its near-metallic band structure still complicates the D3 parametrisation compared to NiO/CoO.
- **$LaMnO_3$ & $SrTiO_3$**: Using the ideal cubic $Pm\bar{3}m$ high-symmetry perovskite phases avoids the complexities of low-temperature distortions (Jahn–Teller distortion in $LaMnO_3$; antiferrodistortive phase in $SrTiO_3$) and provides a cleaner strain sweep for E_{dis}.

Conclusion

This analysis successfully demonstrates a robust Python-based protocol for quantifying D3-induced dispersion errors across diverse oxide frameworks. By utilizing high-symmetry Materials Project IDs ($Fm\bar{3}m$, $P4_2/mnm$, and $Pm\bar{3}m$) as idealized baselines, we isolated the structural contribution to the dispersion energy, revealing that $|\Delta E|$ is highly sensitive to both the unit cell volume and the electronic character of the metal cation.

The observed systematic overestimation of binding in metallic rutile phases compared to insulating perovskite and rock-salt systems confirms the inherent limitations of the atom-pairwise D3 model in itinerant electron environments. Furthermore, the use of ideal cubic phases allowed for the characterization of E_{dis} without the confounding effects of local distortions like the Jahn-Teller effect in $LaMnO_3$. Overall, the `local_max_drop` methodology provides a reliable metric for evaluating semi-empirical dispersion corrections, facilitating more accurate structural predictions in computational materials science.